

No Matrik									

DC014
Sains Komputer 1
Semester I
Sesi 2024/ 2025
2 jam

Name: _____

Group Tutorial: _____

KOLEJ MATRIKULASI PULAU PINANG
KEMENTERIAN PENDIDIKAN MALAYSIA
PENANG MATRICULATION COLLEGE
MINISTRY OF EDUCATION MALAYSIA

PRE-PSPM 1
PROGRAM MATRIKULASI SISTEM EMPAT SEMESTER
PRE -PSPM 1
MATRICULATION PROGRAMME FOUR SEMESTER SYSTEM

SAINS KOMPUTER 1
2 JAM

JANGAN BUKA SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

ARAHAN KEPADA CALON:

Kertas ini mengandungi 6 soalan.
Jawab **semua** soalan di ruang yang disediakan.

INSTRUCTION TO CANDIDATE:

This booklet consists of 6 questions. Answer **all** questions in the space provided.

Untuk Kegunaan Pemeriksa		
No. Soalan	Pemeriksa	KKP/KP
1		
2		
3		
4		
5		
6		
JUMLAH		

1. (a) Maria is a cake decorator who is creating an interactive, multimedia-based learning module to teach her clients how to decorate cakes with different techniques. She plans to make the learning experience enjoyable.

- (i) Identify and describe the most suitable multimedia element Maria should include to make her module engaging for her clients.

[2 marks]

- (ii) Explain the purpose of the multimedia element mentioned in 1(a)(i).

[2 marks]

(b)



Figure 1

- (i) Maria designed a logo for her multimedia-based learning module. Identify and describe the dimension of the multimedia representation in Figure 1.

[2 marks]

2. (a) The design and development process of a multimedia product is crucial to achieving high-quality outcomes.

- (i) Describe Planning in Pre-Production phase.

[2 marks]

- (ii) Based on the description below, identify the multimedia development phases. [4 marks]

Description	Multimedia Phases
I. Jom Sihat team is developing a health-awareness multimedia project. They need to organize content, staff, and resources.	
II. Yusof builds a 3D animated character into an interactive story.	
III. Yasir compiled multimedia elements into an interactive training module.	
IV. Yanna conducts several testing to ensure it's compatible with multiple devices.	

- (b) Identify and describe the type of authoring tool based on scenarios given.

- (i) Munir needs to create an interactive e-learning module that includes multiple drag-and-drop options, quizzes, and educational graphics. [2 marks]

- (ii) Munirah is developing a multimedia project with sound effects synced to specific moments in the animation. [2 marks]

- (c)



Figure 2

- (i) Navigation on a mobile phone enables users to access and move between app features and content with simple, touch-friendly controls. Identify and describe the type of navigation in Figure 2.

[2 marks]

- (d) A teacher wants to present a lesson on ecosystems to her class. She has two options: using a pre-recorded video that can be played from a USB (pendrive) or streaming a live webinar with an expert in the field.

Differentiate the multimedia delivery method according to given criteria.

[4 marks]

	USB	Live Webinar
Ease of editing		
Accessibility		

3. Kamal is a data manager at a retail company, tasked with upgrading their data storage system. Currently, the company is facing the issues of duplicate data.

- (a) Suggest the most appropriate data storage approach to be considered to solve these issues and justify your answer.

[2 marks]

- (b) Describe one (1) advantage and one (1) disadvantage that Kamal encountered after he upgraded to the data storage approach mentioned in 3(a).

[4 marks]

Advantage	Disadvantage

4. Recognizing the critical role of data, Nurul realized that ensuring data integrity is essential.

[2 marks]

(a) Explain the importance of data integrity in databases.

(b) Nurul needs to retrieve specific information about products from the database. Explain one (1) function of the query object.

[2 marks]

5. (a) To streamline teamwork, Sophie wants to set up a working environment that allows team members to work together in real-time, share files, communicate efficiently, and access project resources securely over the network.

(i) Identify the network usage mentioned in the above scenario.

[1 marks]

(ii) State one (1) network usage that Sophie should consider to facilitate virtual meetings.

[1 mark]

(iii) Describe your answer in 5(a)(ii).

[2 marks]

(b) Zara is setting up a network for a new office. Her goal is to ensure that the network supports both wired and wireless connections, allowing users flexibility in accessing resources.

(i) Describe the difference of wired and wireless network media.

[2 marks]

Wired	Wireless

- (ii) Identify one (1) advantage and one (1) disadvantage if Zara considers using fiber-optic cable for the office network setup. [2 marks]

Advantage: _____

Disadvantage: _____

- (iii) Recommend the most cost-effective cable that Zara should consider for her office network. [1 mark]

- (iv) Describe the cable mentioned in answer 5(b)(ii). [2 marks]

- (v) Identify the network medium for remote control in small to medium rooms. [1 mark]

- (c) Diagram 1 below shows a basic network diagram in Emma's office.

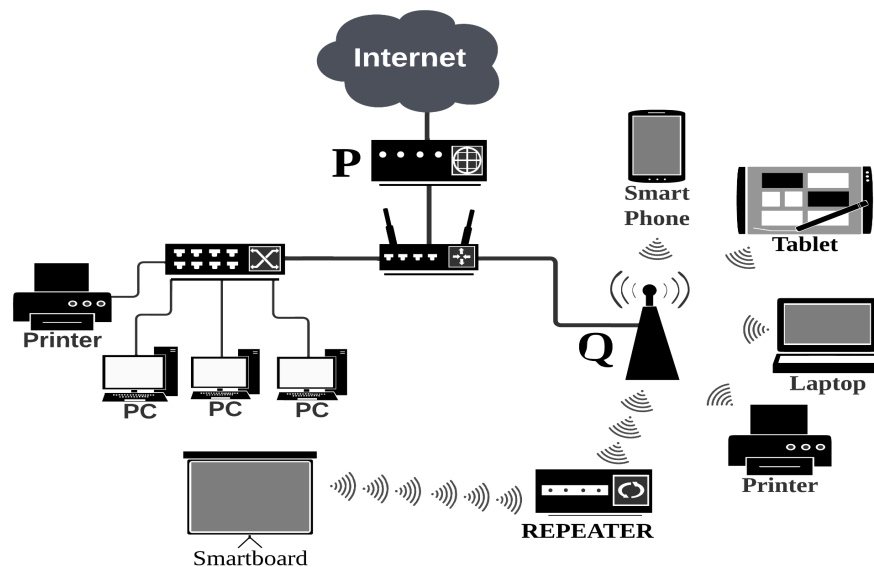


Diagram 1

- (i) Emma's office network requires a **P** to establish a connection with the internet service provider. Name and describe the function of **P**. [2 mark]

- (ii) Emma includes a **Q** in the setup to enable mobile devices access the network. Identify and describe the function of **Q**. [2 marks]

- (iii) Describe the function of a repeater in an office network in a large office area. [2 marks]

6. (a) David is conducting a general study on historical events for a history class. He needs to find reliable resources but has a broad topic and isn't sure what specific events to focus on.

- (i) Identify the most suitable type of search tool David should use to get an overview of historical events. [1 mark]

- (ii) Describe the type of search tool mentioned above. [2 marks]

- (b) Search strategies will help you to narrow down the list of hits to more manageable numbers so that the searching process can be more efficient and effective. Identify the search strategy used based on the scenarios.

- (i) Simply typing "climate change effects" yields too many broad results, so you decide to search for "climate change impact on coastal cities." [1 mark]

- (ii) Typing **site: gov** along with your keyword search limits the hits to government sites. [1 mark]

- (iii) To refine the search further, you decide to narrow it down to cities specifically affected by both climate change and sea-level rise. [1 mark]

(c)



Figure 3

- (i) Identify and describe the web activity in Figure 3.

[3 marks]



Figure 4

- (ii) Identify web activity in Figure 4.

[1 marks]

- (iii) List two (2) examples of web activity in Figure 4.

[2 marks]

- (d) Joe has been researching new laptops online. A few days later, he starts seeing ads for laptops on every website he visits. This makes him uncomfortable, as he feels his online activity is being tracked.

(i) Describe web browsing privacy.

[2 marks]

(ii) Identify the threat he's facing.

[1 mark]

(iii) Suggest one (1) precaution Joe could take to protect his web browsing privacy.

[1 mark]

- (e) After a trip abroad, Lando notices unusual login attempts on his email account from unfamiliar locations. He worries that someone may have gained his email password illegally and logged into his email account.

(i) Identify the threat Lando is facing.

[1 mark]

(ii) Suggest one (1) safeguard he could implement to protect his email account.

[1 mark]

- (f) Mohsin uses a connected car app that monitors his car's fuel level, tire pressure, and maintenance schedule. The app alerts him when it's time for a check-up, helping him keep his vehicle in good condition.

Discuss one benefit of IoT in Mohsin's car.

[2 marks]

END OF QUESTIONS